V. Genetic Disorders

Notes

- A genetic disease is any disease that is caused by an abnormality in an individual genome.
- Some genetic disorders are inherited from the parents, while other genetic disorders are caused by acquired changes or mutations in pre-existing genes or group of genes.
- Mutations can occur either randomly or due to some environmental exposure.
- Some genetic disorders in man are Albinism, Haemophilia, Muscular Dystrophy, Phenylketonuria, Alkaptonuria, etc. These genetic disorders are transferred from one generation to the next generation.
- Down syndrome, Klinefelter syndrome, Turner syndrome are such genetic disorders which are caused due to chromosomal abberations.

Sex chromosomes and sex-linked inheritance :

- Sex chromosome, either of a pair of chromosomes that determine whether an individual is male or female.
- Besides sex-determining gene, there are some other genes, which are found on the sex chromosomes, determining the body character of an individual. Such characters are termed as a **sex-linked character** and its inheritance is called **sex-linked inheritance**.

Albinism :

- Albinism is a congenital genetic disorder characterized in human by the complete or partial absence of pigment **(melanin)** in the skin, hair and eyes.
- Lack of skin pigmentation makes to more susceptibility to **sunburn** and **skin cancer**.
- This also affects essential granules present in immune cells leading to increased susceptibility to infection.
- Albinism results from inheritance of recessive gene alleles.
- It is due to absence or defect of **tyrosinase,** a copper - containing enzyme involved in the production of **melanin**.

Bubble Baby Disease :

- Bubble Baby Disease is also known as severe combined immunodeficiency (SCID), alymphocytosis, Glanzmann-Rinker syndrome, severe mixed immunodeficiency syndrome and thymic alymphoplasia.
- In this disease, children are born without a functioning immune system and in the past were protected from germs within the sterile environment of a plastic bubble.
- This is a rare genetic disorder characterized by the disturbed development of functional T cells and B cells, caused by numerous genetic mutations.
- The only cure currently and routinely available for SCID is bone marrow transplant which provides a new immune system to the patient. Gene therapy treatment of SCID has also been successful in clinical trails but not without complications.

Phenylketonuria :

- Phenylketonuria (PKU) is a type of amino acid metabolism disorder.
- It is an inherited disease.
- In this genetic disorder, due to the lack of **phenylalanine hydroxylase enzyme**, the amino acid phenylalanine is not converted into **tyrosine**. It results in increasing the amount of phenylalanine in the body.
- It can damage the brain and cause severe intellectual disability and mental disorders.

Haemophilia :

- Haemophilia is an inherited genetic condition, meaning it is passed down through families. It impairs the body's ability to make blood clots, a process needed to stop bleeding.
- It is caused by a defect in a gene that determines the formation of factors responsible for blood clotting.
- These genes are located on the X-chromosome, making haemophilia an X-linked recessive disease.
- This disease is generally found in male while the female is the vector of disease.
- Haemophilia is not contagious.
- Haemophilia is also known as **Bleeder's Disease**.
- Haemophilia is also known as **Royal Disease** because the Queen of England **Victoria** was suffering from this disease.

Alkaptonuria :

- **Alkaptonuria** (AKU) is a rare disorder of autosomal recessive inheritance. In this disease the body cannot process the amino acids phenylalanine and tyrosine, which occur in protein.
- It is caused by a mutation in the HGD gene that results in the accumulation of **homogentisic acid (HGA)**.
- Characteristically, the excess HGA means sufferers pass dark urine, which upon standing turns black.
- Over time patients develop other manifestations of AKU due to deposition of HGA in collagenous tissue, namely **ochronosis** (bluish/Black discoloration of tissue) and **ochronotic osteoarthropathy**.

Muscular Dystrophy :

- Muscular dystrophy is caused by an X-linked recessive gene.
- In muscular dystrophy, abnormal genes interfere with the production of proteins needed to form healthy muscle.
- It is an inheritable disease.

Sickle Cell Disease :

- Sickle cell disease (SCD) – also called sickle cell anaemia – is a group of inherited disorders that affect haemoglobin, the major protein that carries oxygen in red blood cells.
- Normally, red blood cells are disc-shaped and flexible so they can move easily through the blood vessels. In sickle cell disease, red blood cells are misshaped, typically crescent-or "sickle"-shaped due to a gene mutation that affects the haemoglobin molecule.
- When red blood cells sickle, they do not bend or move easily and can block blood flow to the rest of the body.

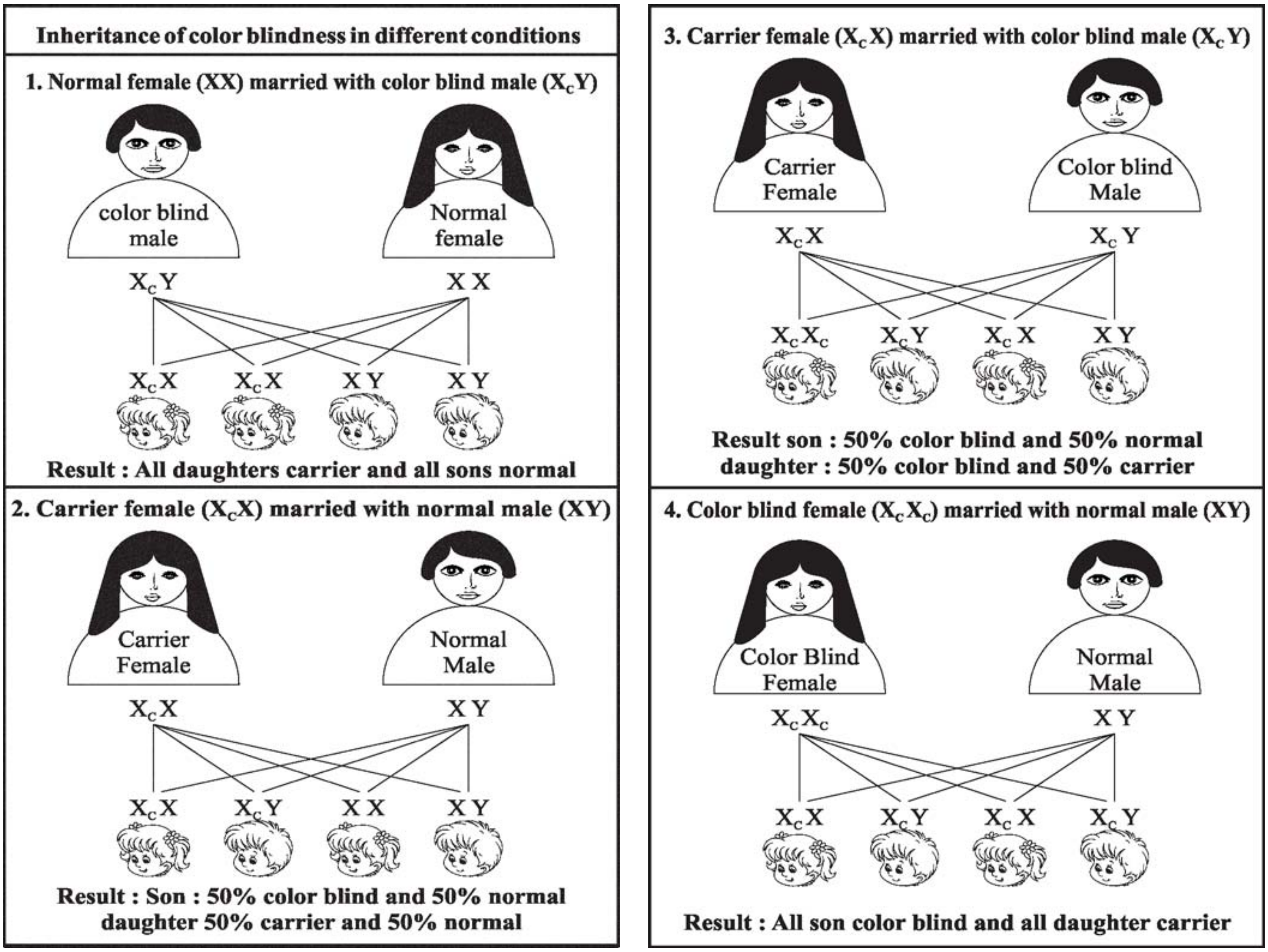
- SCD is a chronic single gene disorder causing a debilitating systemic syndrome characterized by chronic anaemia, acute painful episodes, organ infarction and chronic organ damage and by a significant reduction in life expectancy.
- People who have sickle cell disease may also experience other serious health complications, such as chronic pain, stroke, lung problems, eye problems, infections, and kidney disease.
- The National Sickle Cell Anaemia Elimination Mission (NSCEM), launched by the PM Narendra Modi in Shahdol, Madhya Pradesh in July, 2023, focuses on addressing the significant health challenges posed by sickle cell disease, particularly among tribal populations of the country.
- This mission aims to eliminate sickle cell genetic transmission by the year 2047 in India.

Thalassemia :

- Thalassemia are inherited blood disorders characterized by abnormal haemoglobin production.
- In this condition, body has fewer red blood corpuscles and less haemoglobin than it should. Often there is mild to severe anaemia.
- Haemoglobin is important because it lets RBC carry oxygen to all parts of the body.
- Thalassemia is really a group of blood problems, not just one.
- For synthesis of haemoglobin, two proteins, alpha & beta are required. Without enough of one or the other, RBC cannot carry oxygen as they should.
- Alpha thalassemia means you lack alpha haemoglobin, with beta thalassemia, you lack beta haemoglobin.

Color Blindness :

- Color blindness (Daltonism), also known as **Color Vision deficiency** is the decreased ability to see color or differences in color.
- The most common cause of color blindness is an inherited problem in the development of one or more of the three sets of color sensing **cones** in the eye. It is a recessive X-linked inheritance disease.
- Males are more likely to be color blind than females, as the genes responsible for the most common forms of color blindness are on the X-chromosome.
- As females have two X-chromosomes, a defect in one is typically compensated for by the other, while males have one X-chromosome.
- The females act as a carrier of this disease.
- There are four possibilities of inheritance of color blindness to progeny.



(i) First probability : Marriage between normal female (XX) and color blind male (XcY).

- In this condition, all the daughter will be carrier while all the son will be normal.

(ii) Second probability : Marriage between carrier female (XcX) and normal male (XY).

- In this case the probability among children are as follows :
  50% color blind son; 50% unaffected son;
  50% carrier daughter; and 50% unaffected daughter

(iii) Third probability : Marriage between carrier female (XcX) and color blind male (XcX).

- In this case, the probability of inheritance among progeny is as follows :
  50% color blind son; 50% unaffected son
  50% color blind daughter; 50% carrier daughter

(iv) Fourth probability : Marriage between colorblind female (XcXc) and normal male (XY).

- In this case the probability of inheritance among progeny is as follows :
  All the son will be color blind while all the daughter will be a carrier.

Chromosomal aberrations :

- Chromosomal aberrations are departures from the normal set of chromosomes. It is a missing, extra, or irregular portion of chromosomal DNA.
- They can refer to changes in the number of sets of chromosomes **(ploidy)**, changes in the number of individual chromosomes **(somy)** or changes in the appearance of individual chromosome through mutation-induced rearrangements.
- They can be associated with a genetic disease or with species differences.

Down Syndrome (45+2 = 47 Chromosomes) :

- Down syndrome is usually caused by an error in cell division called **'non-disjunction'**.
- Non-disjunction results in an embryo with three copies of chromosome 21 instead of usual two.
- Prior to or at conception, a pair of the 21st chromosome in either the sperm or the egg fails to separate.
- As the embryo develops, the extra chromosome is replicated in every cell of the body.
- This type of Down Syndrome is called **trisomy 21**.
- At birth, babies with the Down syndrome usually have certain characteristic signs, including flat facial features, small head and ears, short neck, bulging tongue, eyes that slant upward, typically shaped ears and poor muscle tone and mild to moderate mental disability.
- Down syndrome is also known as **Mongoloid idiocy**.

Turner Syndrome (44 + X = 45 Chromosomes) :

- Turner syndrome, a condition that affects only females, results when one of the X chromosomes (sex chromosomes) is partly or completely missing.
- Turner syndrome can cause a variety of medical and developmental problems, including short height, webbed neck, low-set ears, failure of the ovaries to develop and heart defects.
- The females are **sterile**.

Klinefelter Syndrome (44 + XXY = 47 chromosomes) :

- In this syndrome, there are three sex (an extra X-chromosome) chromosomes instead of two sex chromosomes.
- The affected male from this syndrome is **sterile**.
- The symptoms of this syndrome include - Larger breast than normal **(gynecomastia)**, less facial and body hair and it comes later, lessmuscle tone and muscle grow slower than usul, longer arms and legs, wider hips.

Question Bank

1. Which one of the following is not an genetic disease?

(a) Night blindness (b) Albinism
(c) Haemophilia (d) Colour blindness

U.P.P.C.S. (Pre) 2017

Ans. (a)

Night blindness is not a genetic disease. The cause of Night blindness is deficiency of vitamin A. Albinism is genetic disease in which the pigments called melanin is partially or completely absent in the skin, hair and eyes. Haemophilia is a genetic sex-linked disease. Colour blindness is also a genetic disease.

2. Phenylketonuria is an example of an inborn error of metabolism. This 'error' refers to :

(a) Hormonal overproduction
(b) Atrophy of endocrine glands
(c) Inherited lack of an enzyme
(d) Non-disjunction

I.A.S. (Pre) 1994

Ans. (c)

Phenylketonuria (PKU) is a disease caused by a metabolic disorder inherited as a recessive trait. It is an inborn error of metabolism that results in decreased metabolism of amino acid phenylalanine. The dominant gene 'P', in this case, codes for the enzyme, phenylalanine hydroxylase, formed in the liver cells. This enzyme catalyzes conversion of phenylalanine to tyrosine. In homozygous recessive genotypes, the absence of this enzyme causes a high level of phenylalanine in blood and tissues fluids. The phenotypic effects include a progressive mental retardation starting a few month after birth, seizures, and anomalies of teeth enamel and bones.

3. A person affected by phenylketonuria disease suffers form :

(a) Kidney failure (b) Liver failure

(c) Mental idiocy (d) Impotence

R.A.S./R.T.S.(Pre) 1999

Ans. (c)

See the explanation of above question.

4. The famous 'Bubble Baby Disease' is so called because :

(a) It is caused by water bubble
(b) The suffering baby makes bubbles of saliva
(c) The suffering baby is treated in a germ-free plastic bubble
(d) It is cured only water bubble

U.P.P.C.S. (Pre) 1997

Ans. (c)

Bubble Baby Disease (severe combined immunodeficiency : SCID) is a rare genetic disorder characterized by the disturbed development of functional T cells and B cells caused by numerous genetic mutations. There are several forms of SCID. The famous 'Bubble Baby Disease' is named so as the suffering baby is treated in a germ-free plastic bubble.

5. Haemophilia is a genetic disease carried by –

(a) Women, appear in women
(b) Women, appear in men
(c) Men, appear in women
(d) Men, appear in men

I.A.S. (Pre) 1993

Ans. (b)

Haemophilia is a genetic sex-linked disease in which blood clotting takes a long time in a patient (up to 24 hours) due to lack of thromboplastin protein in blood plasma. Women carry it while it appears in men (because men have one X-chromosome but women have two). It is also called bleeder's disease which was first found in Queen Victoria. So it is also known as 'royal haemophilia' or 'the royal disease'.

6. Haemophilia is :

(a) Caused by bacteria (b) Caused by virus
(c) Caused by pollutants (d) A hereditary defect

U.P. Lower Sub. (Pre) 2015

Ans. (d)

See the explanation of above question.

7. Which one of the following genetic diseases is sex-linked?

(a) Royal haemophilia (b) Tay-Sachs disease
(c) Cystic fibrosis (d) Hypertension

U.P.P.C.S. (Spl.) (Pre) 2008
I.A.S. (Pre) 1999

Ans. (a)

See the explanation of above question.

8. Haemophilia is a hereditary disease which affects as –

(a) Lack of Hb (b) Rheumatoid leant disease
(c) Lack in WBC (d) Absence of blood clotting

U.P.P.C.S. (Pre) 2003

Ans. (d)

See explanation of above question.

9. The heritable disease is :

(a) Haemophilia (b) T.B.
(c) Cancer (d) Jaundice

U.P.P.C.S. (Mains) 2007
Uttarakhand P.C.S. (Pre) 2006

Ans. (a)

See the explanation of above question.

10. Which of the following is not a genetic disorder?

(a) Down syndrome
(b) Haemophilia
(c) Irritable Bowel Syndrome (IBS)
(d) Sickle Cell Anaemia

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

Down syndrome, haemophilia and sickle cell anaemia are hereditary diseases. Irritable Bowel Syndrome (IBS) is not a genetic disorder, it is a common gut disorder. The cause is not well known. Symptoms can be quite variable and include tummy (abdominal) pain, bloating and sometimes bouts of diarrhoea and/or constipation.

11. In humans trisomy of chromosome number 21 is responsible for :

(a) Haemophilia (b) Klinefelter Syndrome
(c) Down Syndrome (d) Turner Syndrome

R.A.S/R.T.S (Pre) 2018

Ans. (c)

Down Syndrome, also known as trisomy 21, is a genetic disorder caused by the presence of all or part of the third copy of chromosome 21. It is usually associated with physical growth delays, mild to moderate intellectual disability, and characteristic facial features. It is also known as Mongolian idiocy.

12. In human beings, disorder caused due to trisomy of 21^{st} chromosome is :

(a) Turner Syndrome
(b) Down Syndrome
(c) Super Female Syndrome
(d) Klinefelter Syndrome

M.P. P.C.S. (Pre) 2024

Ans. (b)

See the explanation of above question.

13. **Trisomy 21 is known as :**
(a) Evans syndrome (b) Edwards syndrome
(c) Down syndrome (d) Gray baby syndrome
R.A.S./ R.T.S. (Pre) 2021

Ans. (c)

See the explanation of above question.

14. **Down syndrome is a genetic disorder, which is caused due to –**
(a) Due to changes in the number of the chromosomes
(b) Due to changes in the structure of the chromosome
(c) Due to changes in the structure of D.N.A.
(d) Due to changes in the structure of R.N.A.
R.A.S./R.T.S. (Pre) 1999

Ans. (a)

Down syndrome is a genetic disorder which is caused due to changes in the number of the chromosomes. In this, 21^{st} pair of the chromosome are 3, instead of 2. So the chromosome group is [$2x$ + 1(chromosome number 21) = 47]. This syndrome is also known as Mongolian idiocy.

15. **Which one of the following is caused by the expression of a recessive gene present on sex chromosome ?**
(a) Rheumatism (b) Nervous shock
(c) Muscular dystrophy (d) Cerebral hemorrhage
I.A.S. (Pre) 1994

Ans. (c)

Muscular dystrophy is a hereditary disease linked with X-chromosome which is derived by the execution of the ineffective genes present on X-chromosome. This moves from generation to generation.

16. **Match List-I (Disease) with List-II (Types of disease) and select the correct answer using the codes given below :**

List-I	List-II
A. Haemophilia	**1. Deficiency disease**
B. Diabetes	**2. Genetic disease**
C. Rickets	**3. Hormonal disorder**
D. Ringworm	**4. Fungal infection**

Code :

	A	B	C	D
(a)	2	3	4	1
(b)	2	3	1	4
(c)	3	2	1	4
(d)	3	2	4	1

I.A.S. (Pre) 2000

Ans. (b)

The correctly matched lists are as follows :

Haemophilia	–	Genetic disease
Diabetes	–	Hormonal disorder
Rickets	–	Deficiency disease
Ringworm	–	Fungal infection

17. **Which one of the following sets is correctly matched ?**
(a) Diphtheria, Pneumonia and Leprosy – Sex-linked
(b) AIDS, Syphilis and Gonorrhoea – Bacterial
(c) Colour blindness, Haemophilia and Sickle cell anaemia – Hereditary
(d) Polio, Japanese B encephalitis and Plague – Viral
I.A.S. (Pre) 1995

Ans. (c)

Diphtheria, pneumonia and leprosy are bacterial diseases. AIDS is a viral disease while syphilis and gonorrhoea are bacterial diseases. Colour blindness, haemophilia and sickle cell anaemia are hereditary diseases. Polio and Japanese B encephalitis are viral diseases, while plague is a bacterial disease.

18. **In respect of women, men can suffer more of colour blindness because-**
(a) They pass more through mental tensions
(b) They live out of home relatively more
(c) They have only one X-chromosome
(d) Commonly they have less fat
41st B.P.S.C. (Pre) 1996

Ans. (c)

Colour blindness is inherited in an X-linked recessive pattern. The genes are located on the X-chromosome which is one of the two sex chromosomes. In males (who have only one X-chromosome) one genetic change in each cell is sufficient to cause the condition. Males are affected by X-linked recessive disorders much more frequently than females (who have two X-chromosomes) in which a genetic change would have to occur on both copies of the chromosome to cause the disorder. A characteristic of X-linked inheritance is that father cannot pass X-linked traits to their sons.

19. **Which one of the following pairs is incorrectly matched?**
(a) Thiamine – Beriberi
(b) Ascorbic acid – Scurvy
(c) Vitamin A – Colour blindness
(d) Vitamin K – Blood clotting
U.P. P.C.S. (Pre) 2016

Ans. (c)

Lack of vitamin A causes night blindness not colour blindness. Colour blindness is a genetic disease which is inherited in an X-linked recessive pattern. Hence, option (c) is not correctly matched.

20. **If a colour blind man marries with normal women, then the symptoms of colour blindness is generated in :**
(a) Sons (b) Daughters
(c) Sons of sons (d) Sons of daughters
U.P.P.C.S. (Pre) 2009

Ans. (d)